Welcome to the Biodiversity Genomics course!

Joana Meier, Karin Näsvall, Nicol Rueda, Fernando Seixas
Tree of Life Programme, Wellcome Sanger Institute

Gustavo Silva
Universidad Nacional de Colombia

THE ROYAL SOCIETY
The Branco Weiss Fellowship Society in Science
wellcome
Tree of Life
wellcome sanger institute
UNIVERSITY OF CAMBRIDGE

Course teachers

Joana Meier

Nicol Rueda

Karin Näsvall

Fernando Seixas

Gustavo Silva

https://github.com/rapidspeciation/biodiversity_genomics_course

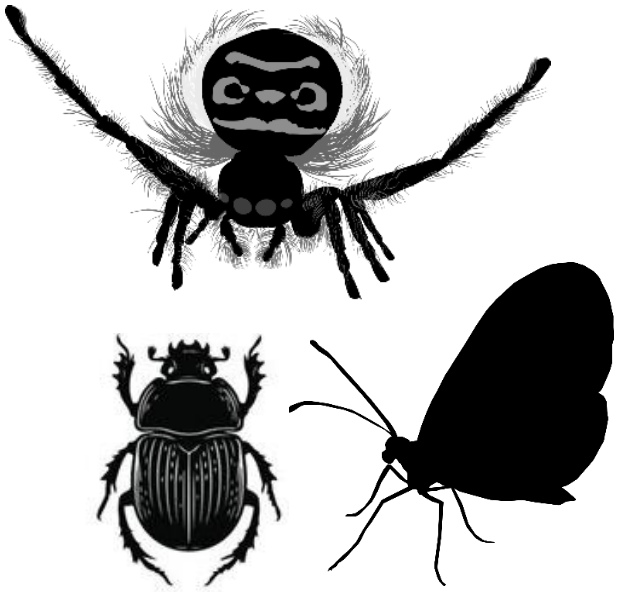
Joana Meier

Childhood

PhD on cichlid fish speciation in Bern, Switzerland

Postdoc at the University of Cambridge, UK

Group leader at the Wellcome Sanger Institute since 2022



Why do some lineages diversify very fast?

Lake Victoria cichlid fishes

500 species

1 species

1 species

1 species

1 species

Meier et al. 2017, Nature Communications
Meier et al. 2023, Science

Nicol Rueda

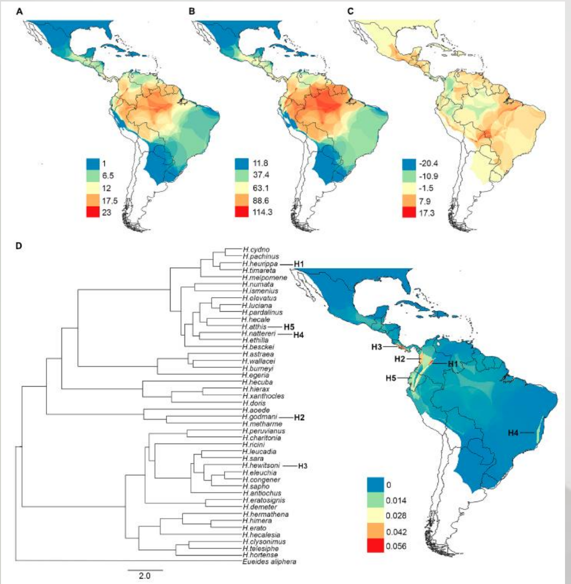
Postdoctoral fellow
Rapid speciation group, Tree of life
Wellcome Sanger Institute

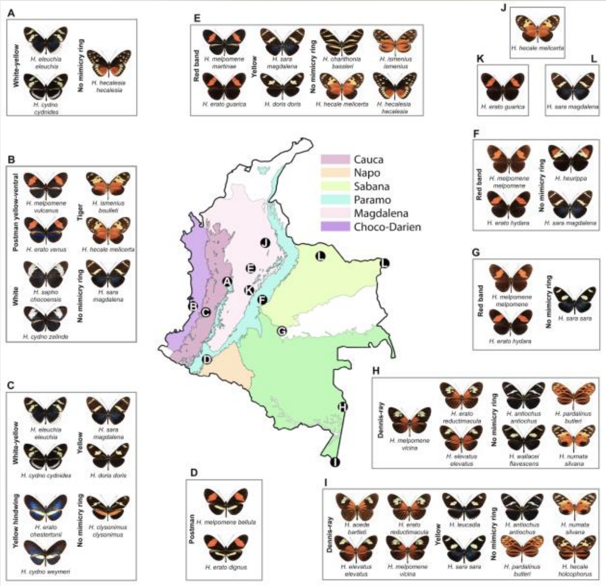
Master of science in Biology in Col/Brazil

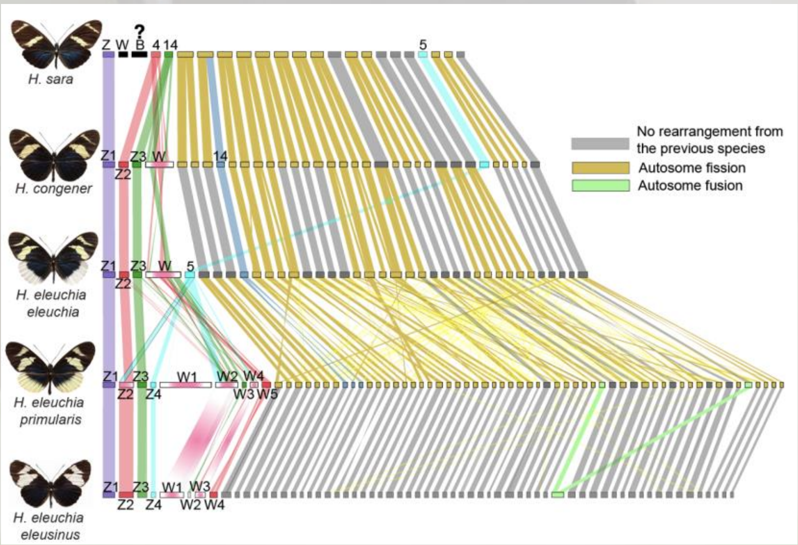
Population monitoring for 7 months in two forests of different conservation status.

Capture-Mark-recapture technique

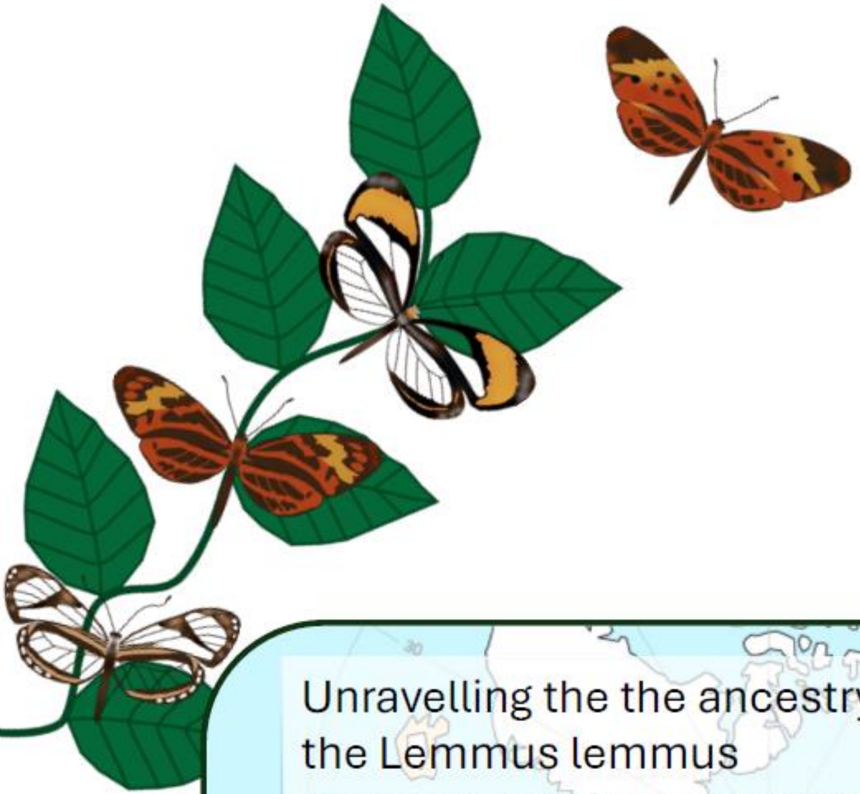
PhD in Biology in Colombia





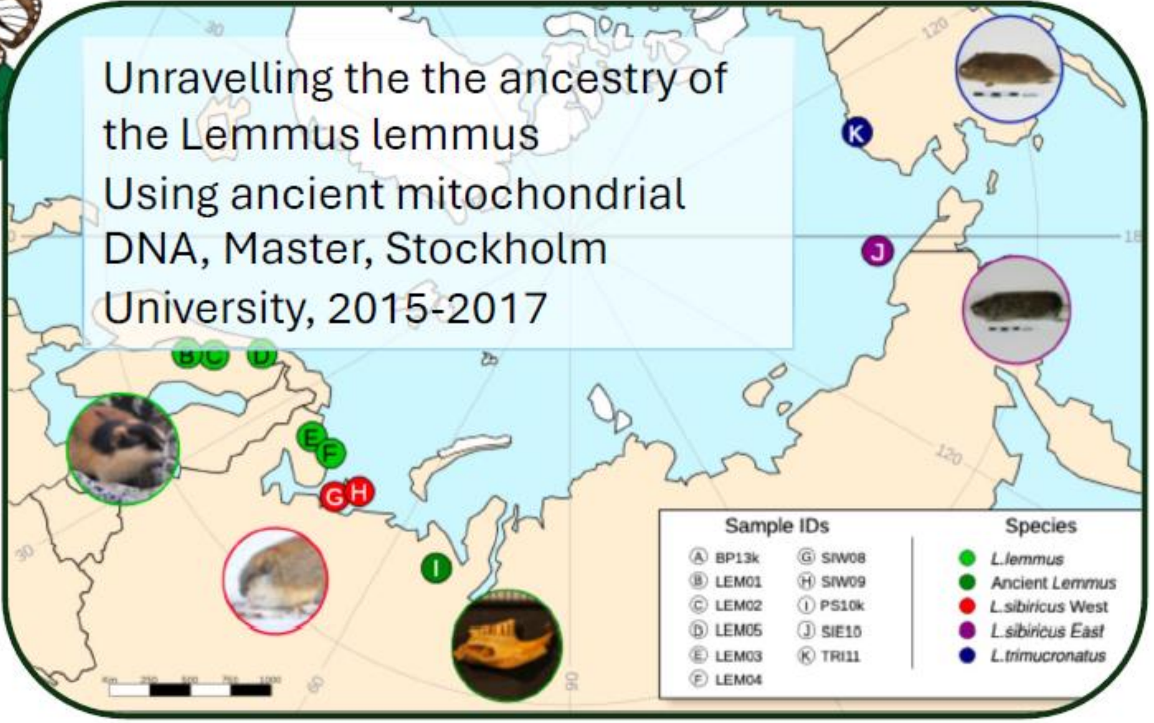


Sex chromosome evolution in Lepidoptera

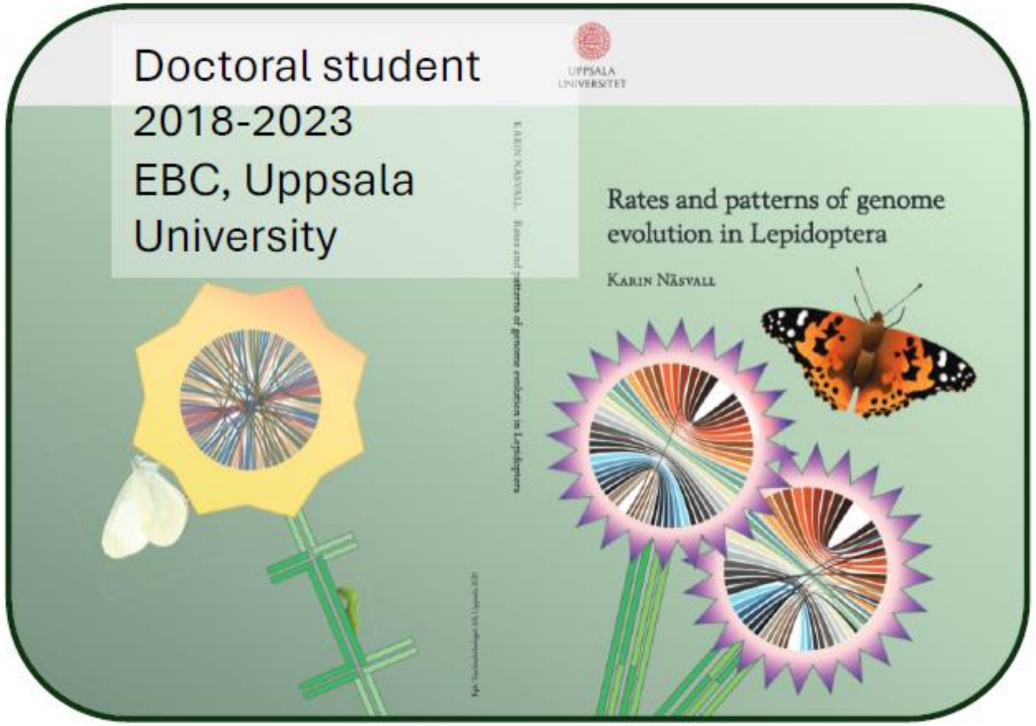


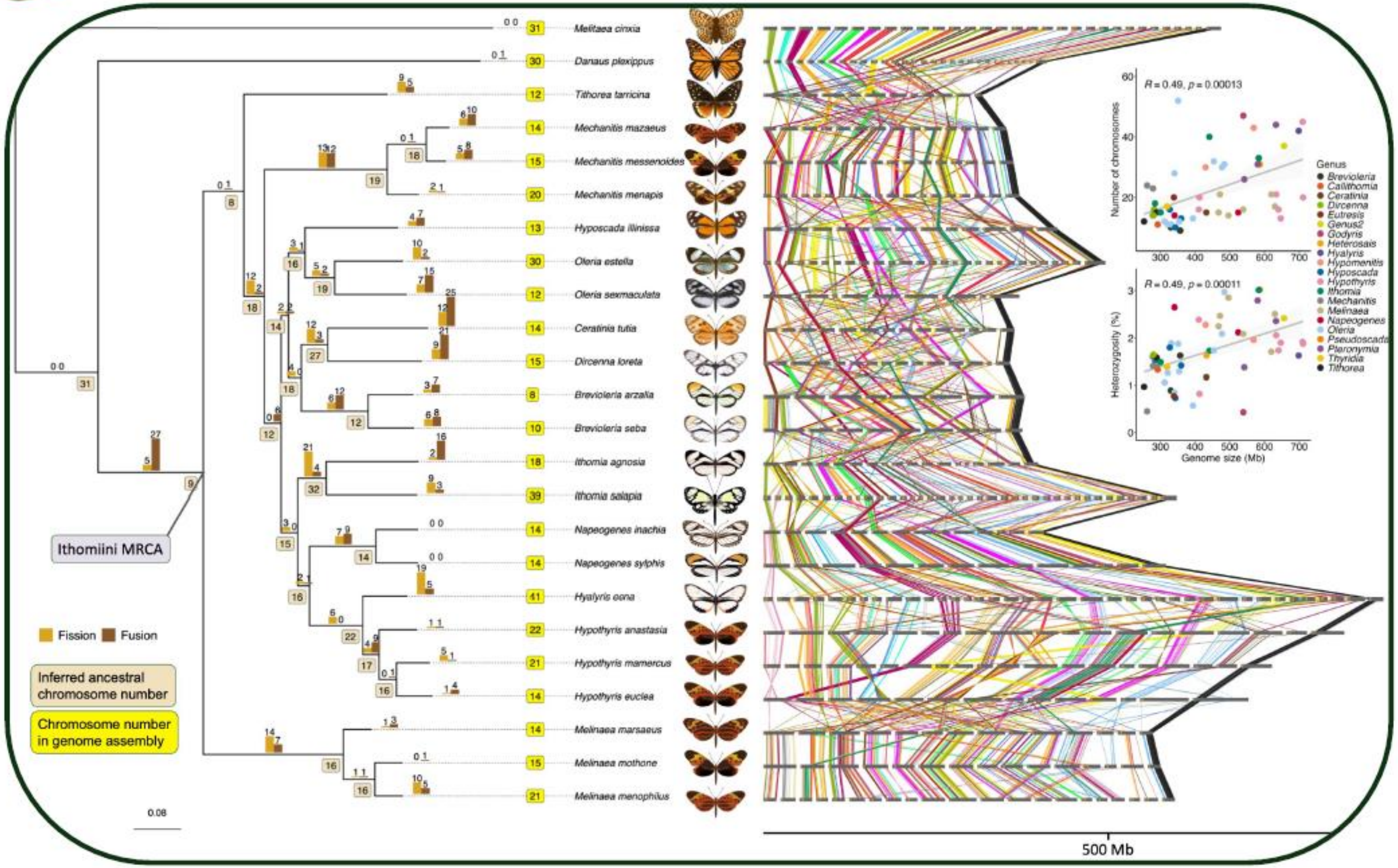
Karin Näsvall, PhD
Postdoctoral fellow
Rapid speciation group, Tree of Life
Wellcome Sanger Institute, UK

Unravelling the the ancestry of the Lemmus lemmus Using ancient mitochondrial DNA, Master, Stockholm University, 2015-2017



Doctoral student 2018-2023 EBC, Uppsala University





Association between speciation rate
And chromosomal rearrangements in Ithomiini butterflies

Fernando Seixas | My timeline

2010 **Bachelor** of Science in Biology

U. PORTO

2013 **Master of Science** in Biodiversity, Genetics and Evolution
(Supervisors: José Melo-Ferreira & Paulo Célio Alves)

CIBIO Centro de Investigação em Biodiversidade e Recursos Genéticos U. PORTO

My timeline

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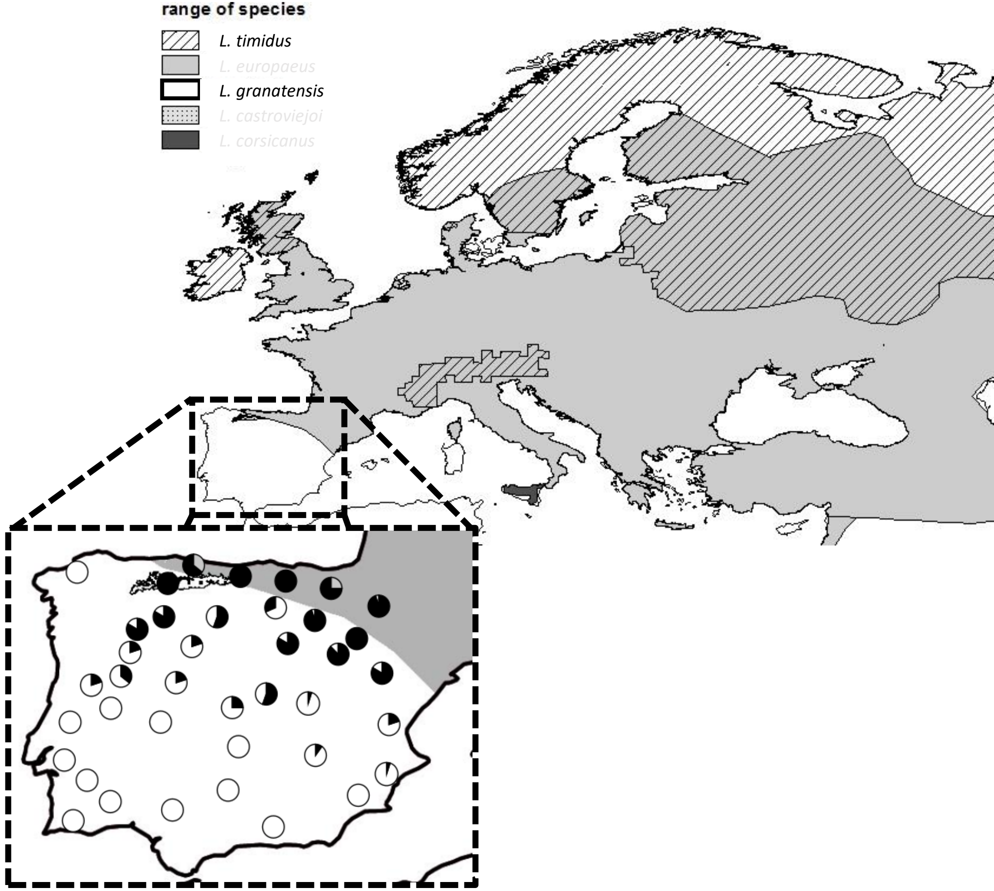

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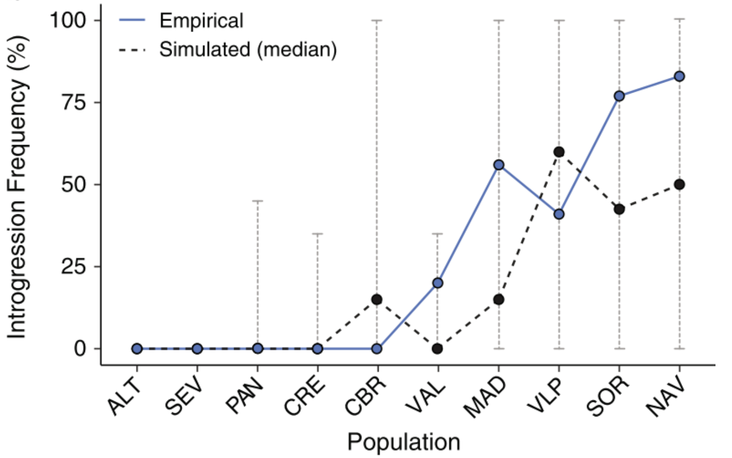

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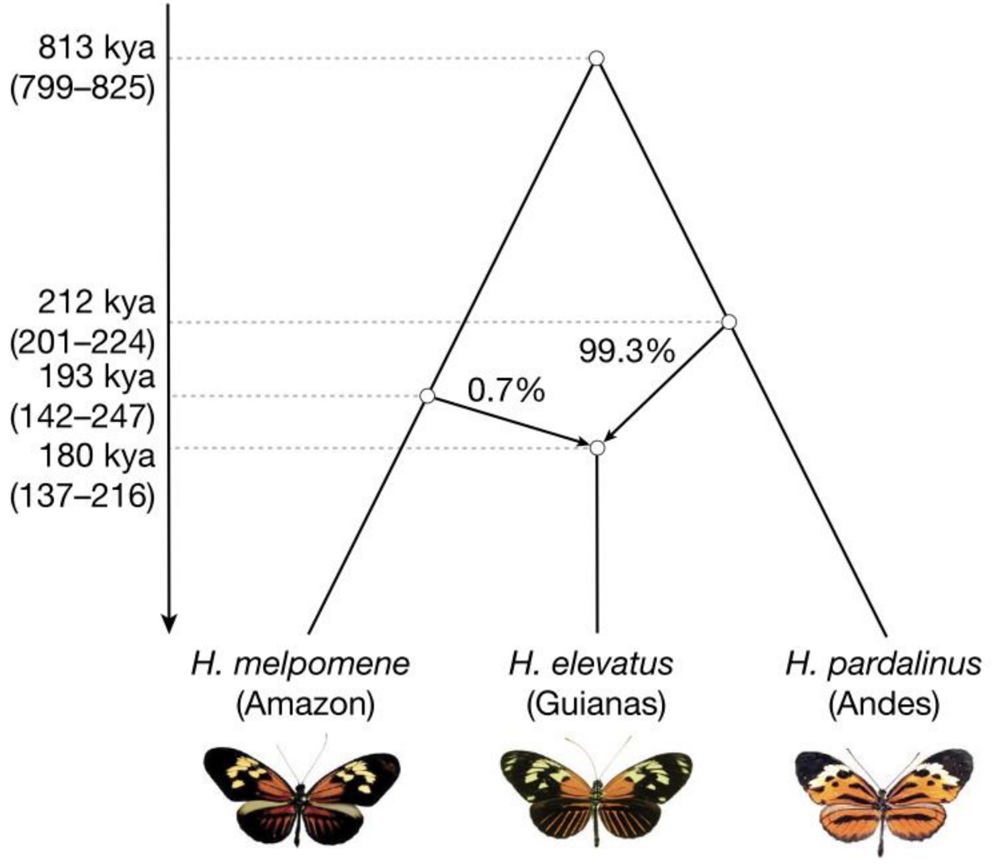
isem Institut des Sciences de l'Evolution-Montpellier; Université Montpellier; CIBIO Centro de Investigação em Biodiversidade e Recursos Genéticos; U. PORTO

2019-2023 **Postdoctoral Fellow**
(Advisor: James Mallet)

HARVARD Faculty of Arts and Sciences
ORGANISMIC AND EVOLUTIONARY BIOLOGY



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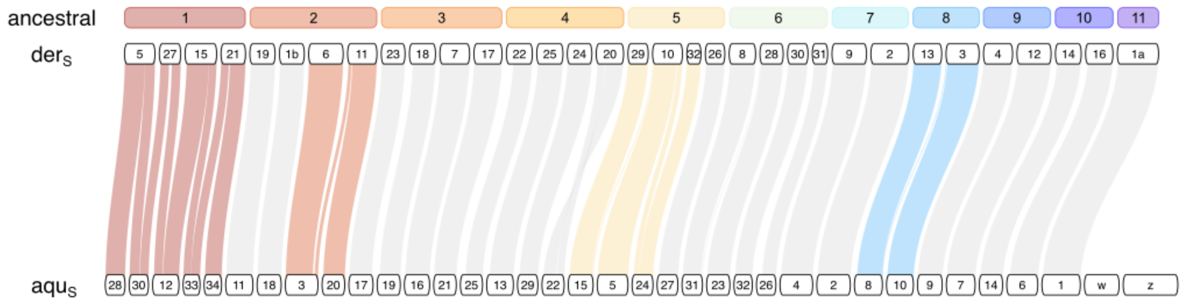
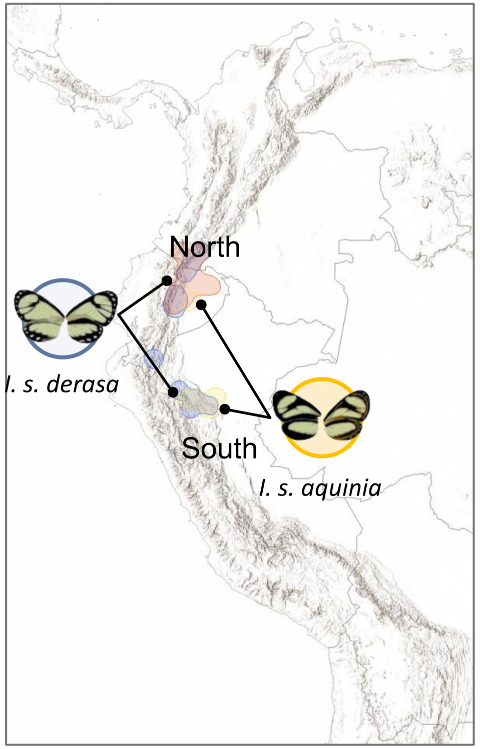
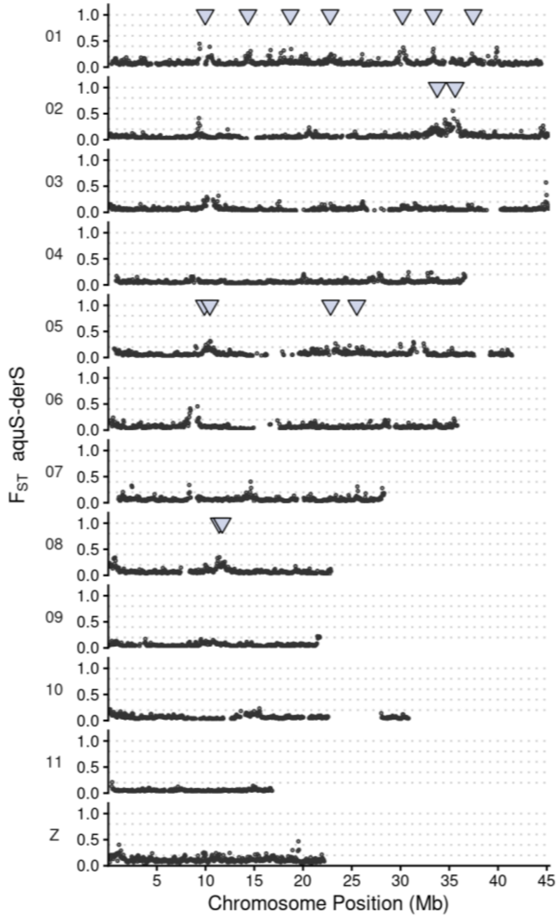
2019-2023 — **Postdoctoral Fellow**
(Advisor: James Mallet)

2024-present — **Postdoctoral Researcher**
(Advisor: Joana Meier)



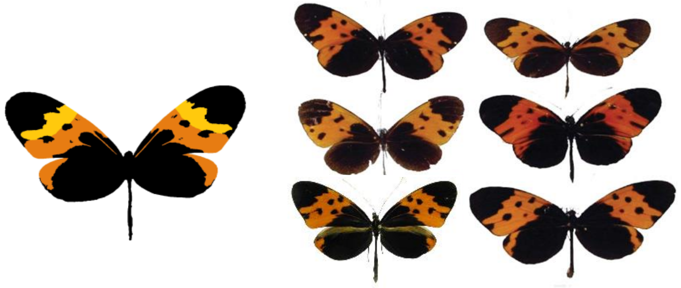




Introduction to
Biodiversity Genomics

Using genomics to understand and preserve biodiversity from genetic diversity, populations, to species and ecosystems

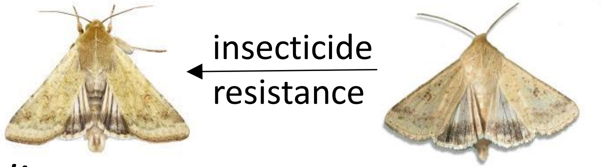
Resolving the taxonomy
- Placing potentially new species
- Species delineation



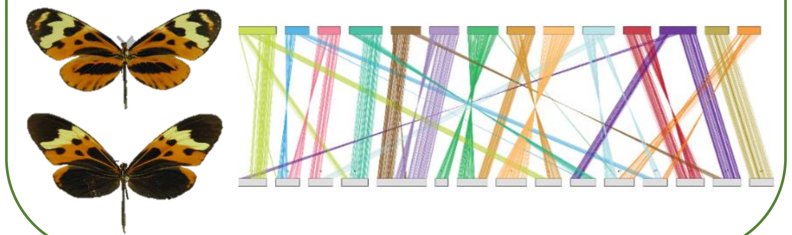
Adaptation and speciation
- Identifying genomic regions involved in speciation
- Identifying genes underlying traits

Informing conservation
- Assessing genetic diversity
- Assessing connectivity

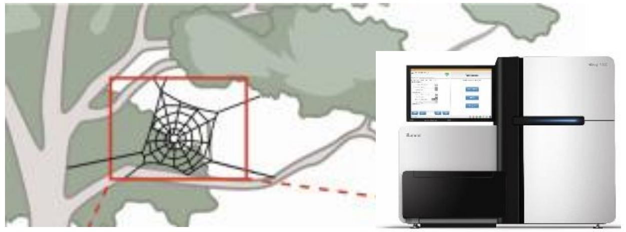
Gene flow
- Are populations/species hybridising or have in the past?
- Finding regions of adaptive introgression



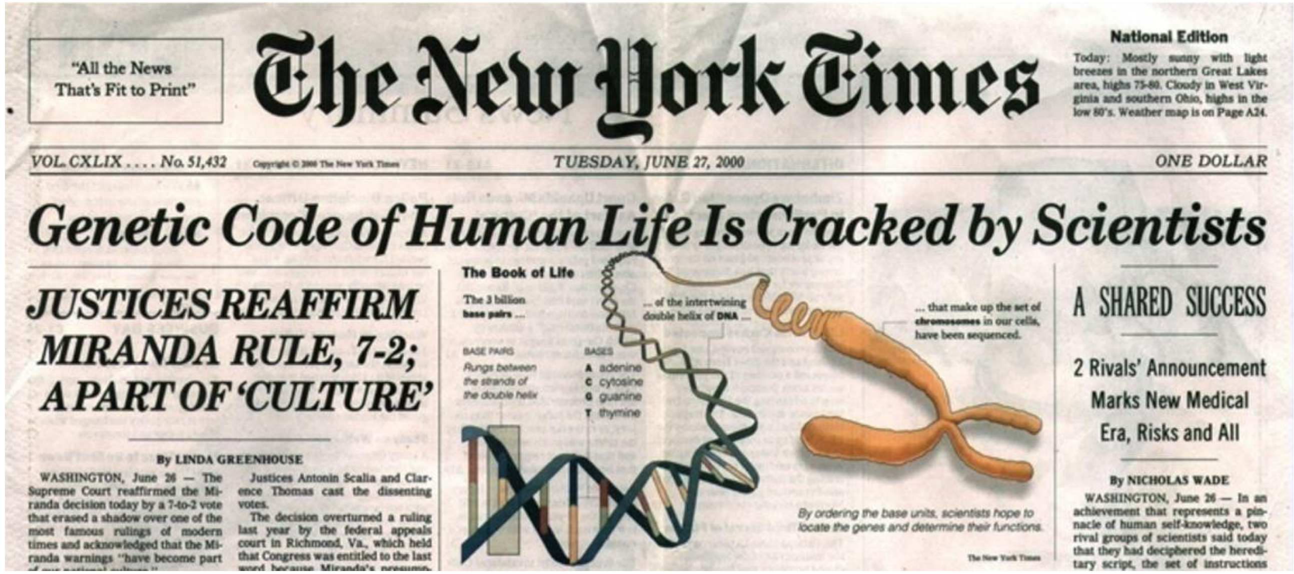
Genome evolution
- Gene expansions, e.g. olfactory
- Chromosomal rearrangements
- Genome size evolution (TEs, etc)



Which species occur here?
- Identifying biodiversity hotspots
- Monitoring effectiveness of conservation strategies



Human Genome Project



"All the News That's Fit to Print"

The New York Times

National Edition

Today: Mostly sunny with light breezes in the northern Great Lakes area, highs 75-80. Cloudy in West Virginia and southern Ohio, highs in the low 80's. Weather map is on Page A24.

VOL. CXLIX .... No. 51,432 Copyright © 2000 The New York Times TUESDAY, JUNE 27, 2000 ONE DOLLAR

Genetic Code of Human Life Is Cracked by Scientists

JUSTICES REAFFIRM MIRANDA RULE, 7-2; A PART OF 'CULTURE'

By LINDA GREENHOUSE

WASHINGTON, June 26 — The Supreme Court reaffirmed the Miranda decision today by a 7-to-2 vote that erased a shadow over one of the most famous rulings of modern times and acknowledged that the Miranda warnings "have become part

Justices Antonin Scalia and Clarence Thomas cast the dissenting votes.

The decision overturned a ruling last year by the federal appeals court in Richmond, Va., which held that Congress was entitled to the last

A SHARED SUCCESS

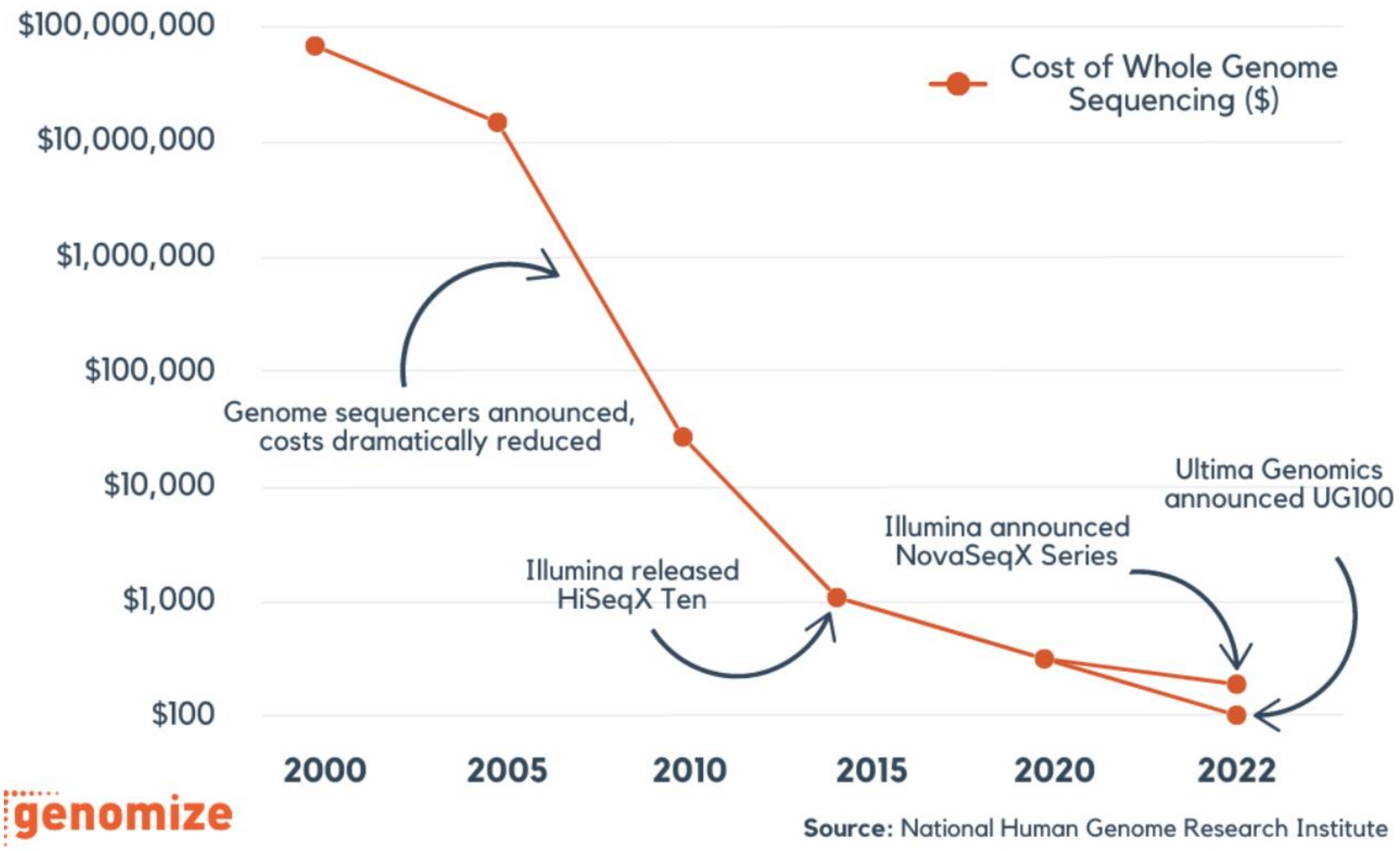
2 Rivals' Announcement Marks New Medical Era, Risks and All

By NICHOLAS WADE

WASHINGTON, June 26 — In an achievement that represents a pinnacle of human self-knowledge, two rival groups of scientists said today that they had deciphered the hereditary script, the set of instructions

- Human genome project – started in 1990, completed in 2003
- Sequenced across ~20 institutions worldwide
- Cost an approximate $5 billion US dollars

Sequencing costs are decreasing rapidly



EARTH BIOGENOME PROJECT

Darwin TREE of LIFE

VGP VERTEBRATE GENOMES PROJECT

A PROJECT OF THE G10K CONSORTIUM

PROJECT PSYCHE

LEPIDOPTERAN GENOMES FOR EUROPE

African BioGenome Project

Genomics for the future of biological diversity across Africa

ATLASea

Atlas des génomes marins

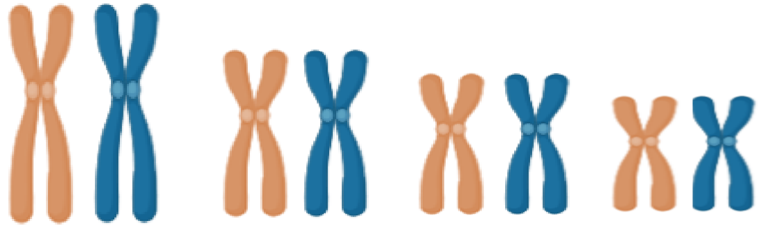
BAT 1K

CREATING A NEW FOUNDATION FOR BIOLOGY

Sequencing Life for the Future of Life

How do we make a reference genome?

Genome = set of all chromosomes



Break into many million
long (10-50 kbp) DNA fragments

Long-read sequencing
machine (e.g. PacBio)

more expensive than
short-read sequencing

Millions of reads,
10-20 kbp long

GATGCTGAGTA
ATAGTGTGGAT
GTGTGGATGTG
TGCTGAGTTCG
TGGATGCTGAT
CTGAGTTCTCG

puzzling them together
(aligning them to
each other)

Reference genome

ATAGTGTGGATGCTGAGTTCGT

```
ATAGTGTGGATG
   GTGTGGATGCT
       TGGATGCTGAGT
          GATGCTGAGTTC
             TGCTGAGTTCG
                CTGAGTTCGT
```

Draft genome

Contig 1

ATAGTGTGGATGCTGAGTTCGT

```
ATAGTGTGGATG
    GTGTGGATGCT
        TGGATGCTGAGT
           GATGCTGAGTTC
              TGCTGAGTTCG
                 CTGAGTTCGT
```

Contig 2

TTCAGGGGCATAAAAGTAACGTA

```
TTCAGGGGCAT
    AGGGGCATAAAA
         GGCATAAAAGTAACG
            CATAAAAGTAACGTA
```

Contig 3

GACACCCGCATATTAGTAAC

```
GACACCCGCAT
    ACCCGCATATTA
        CGCATATTAGTA
           CATATTAGTAAC
```

Adding Hi-C data

Contig 1

ATAGTGTGGATGCTGAGTTCGT

Contig 2

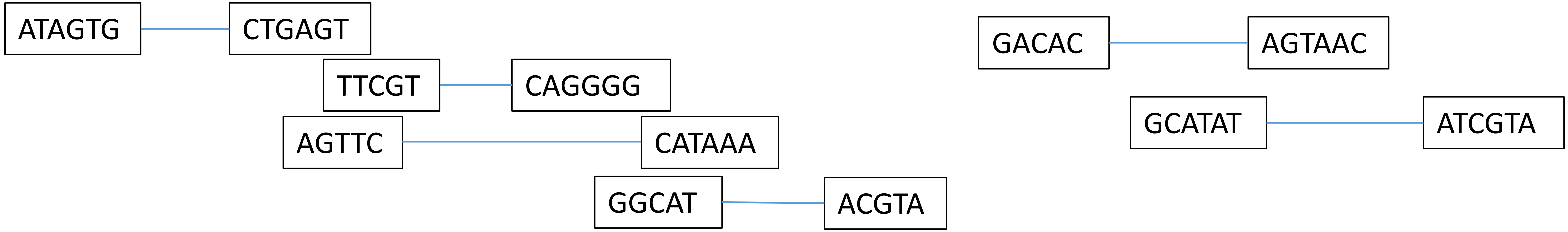
TTCAGGGGCATAAAAGTAACGTA

Contig 3

GACACCCGCATATTAGTAAC

Contig 4

ATCGTA...



Scaffold 2

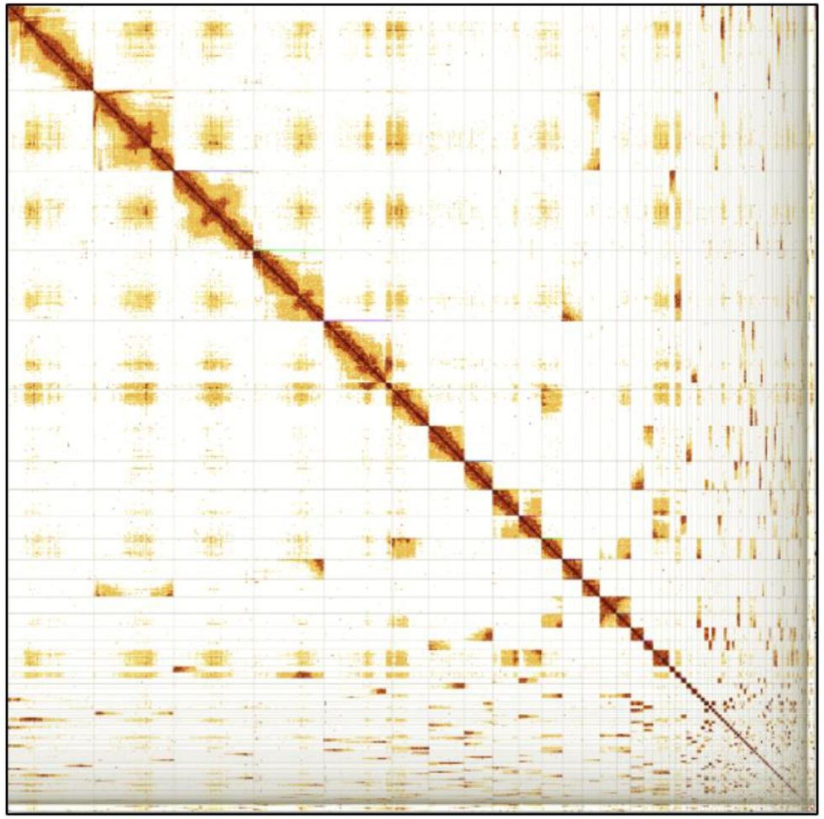
ATAGTGTGGATGCTGAGTTCGT **NNN** TTCAGGGGCATAAAAGTAACGTA

Scaffold 2

GACACCCGCATATTAGTAAC **NNN** ATCGTA...

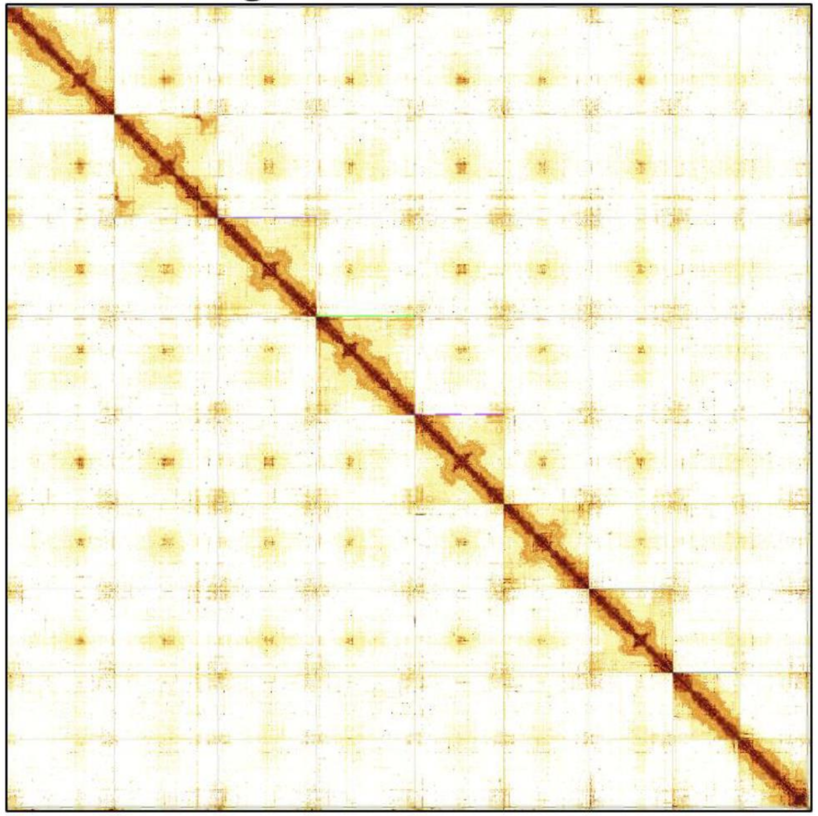
Manual curation

n = 230
N50 = 33.1Mb



Pre curation assembly

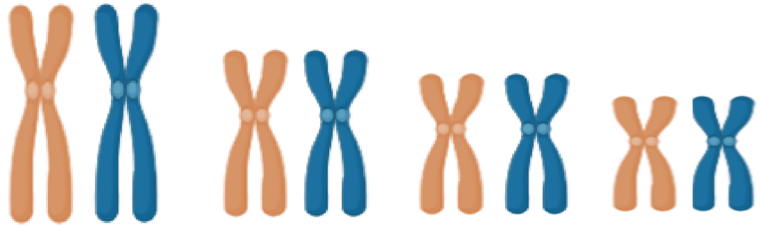
n = 62
N50 = 87.1Mb
99.85% of genome in 9 Chromosomes



Post curation

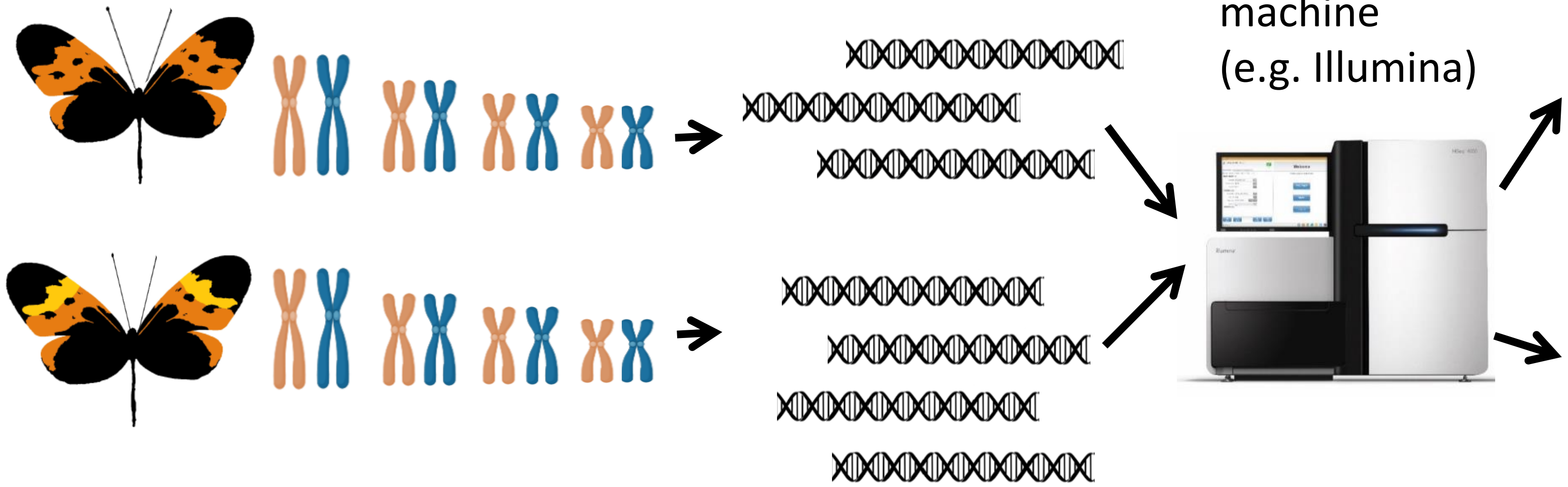
Whole genome resequencing with short reads

Genome = set of all chromosomes



Break into many million short DNA fragments

Short-read sequencing machine (e.g. Illumina)



GATGCT
ATAGTG
GTGTGG

GTGTAG
GATGCT
CTGAGT
TGCTGA

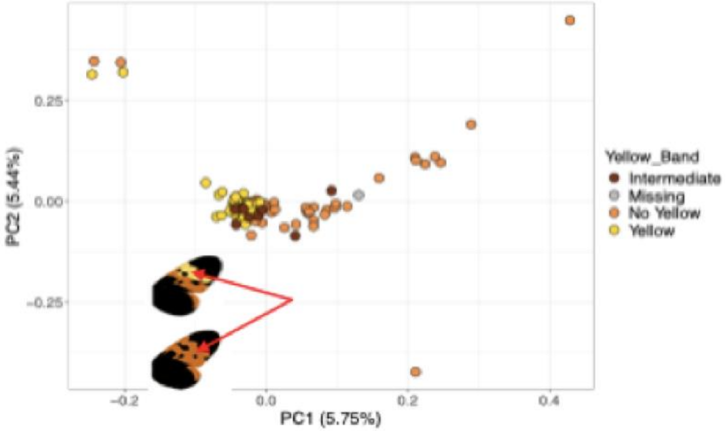
Reference genome

```
ATAGTGTGGATGCTGAGTTCGT
         GATGCT
ATAGTG
   GTGTGG

   GTGTAG
         GATGCT
              CTGAGT
           TGCTGA
```

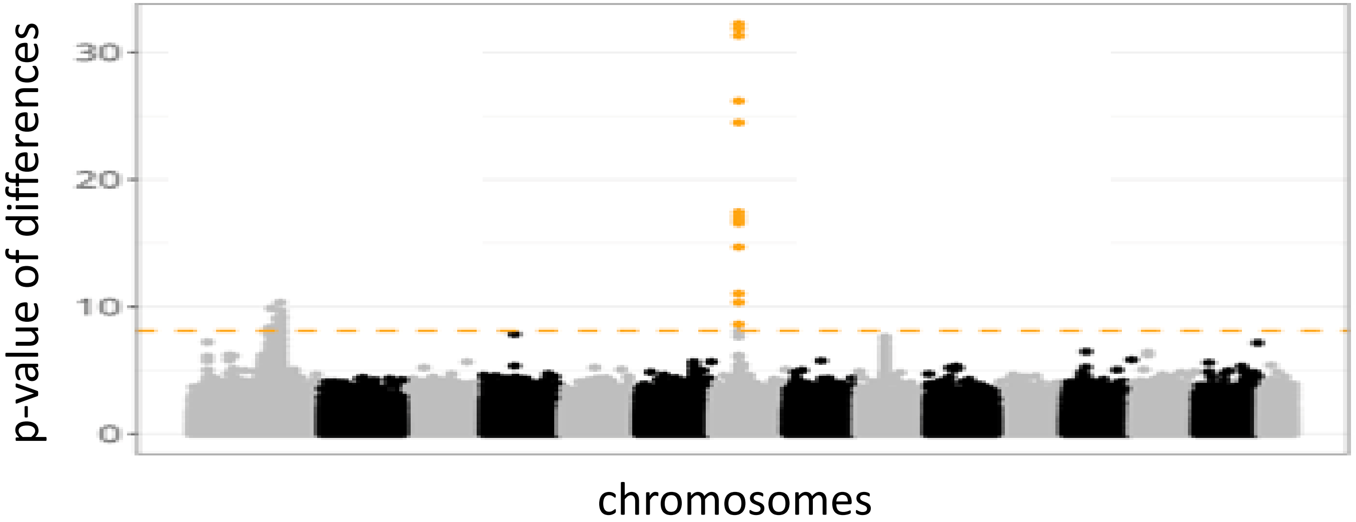
Do these two butterflies belong to different species?

Eva van der Heijden



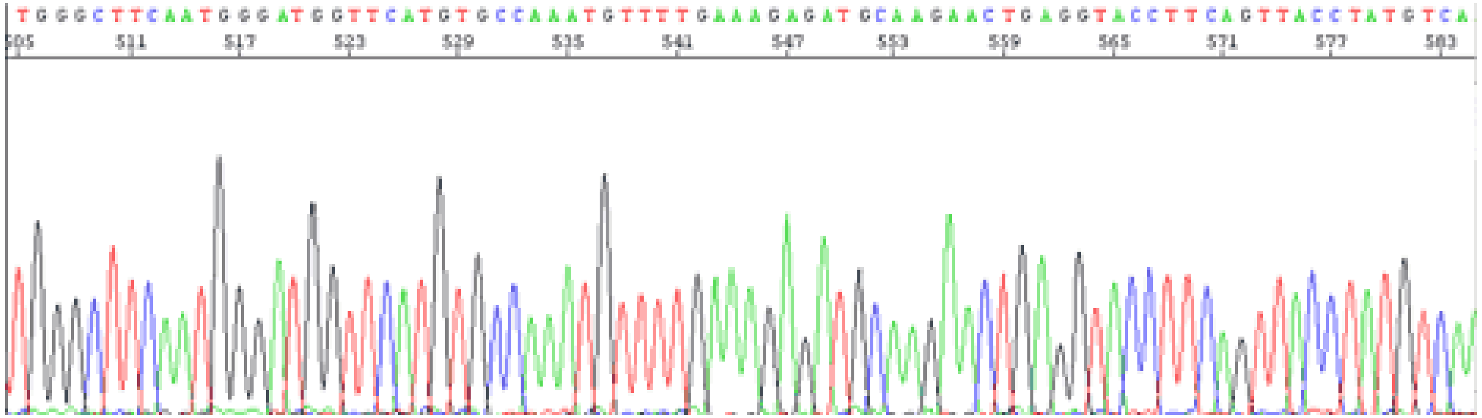
CRISPR butterflies with *ivory* knocked-out

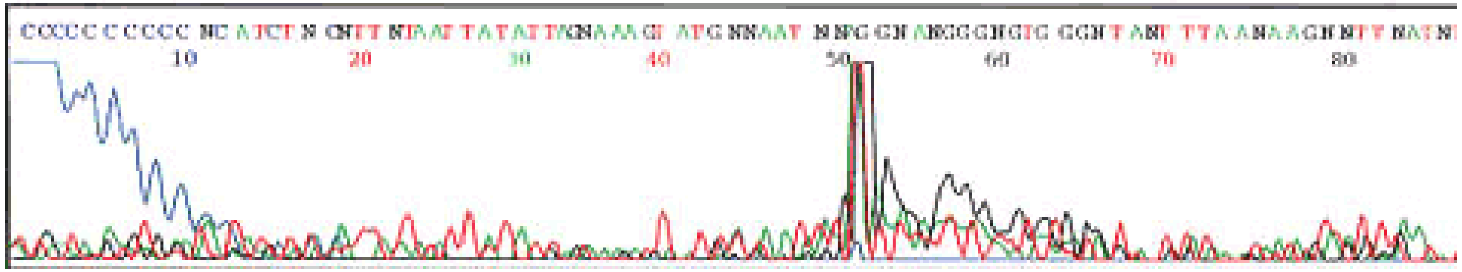
They differ at only one region in the genome



Introduction to high-throughput or next-generation sequencing

Sanger Sequencing (since 1980s)





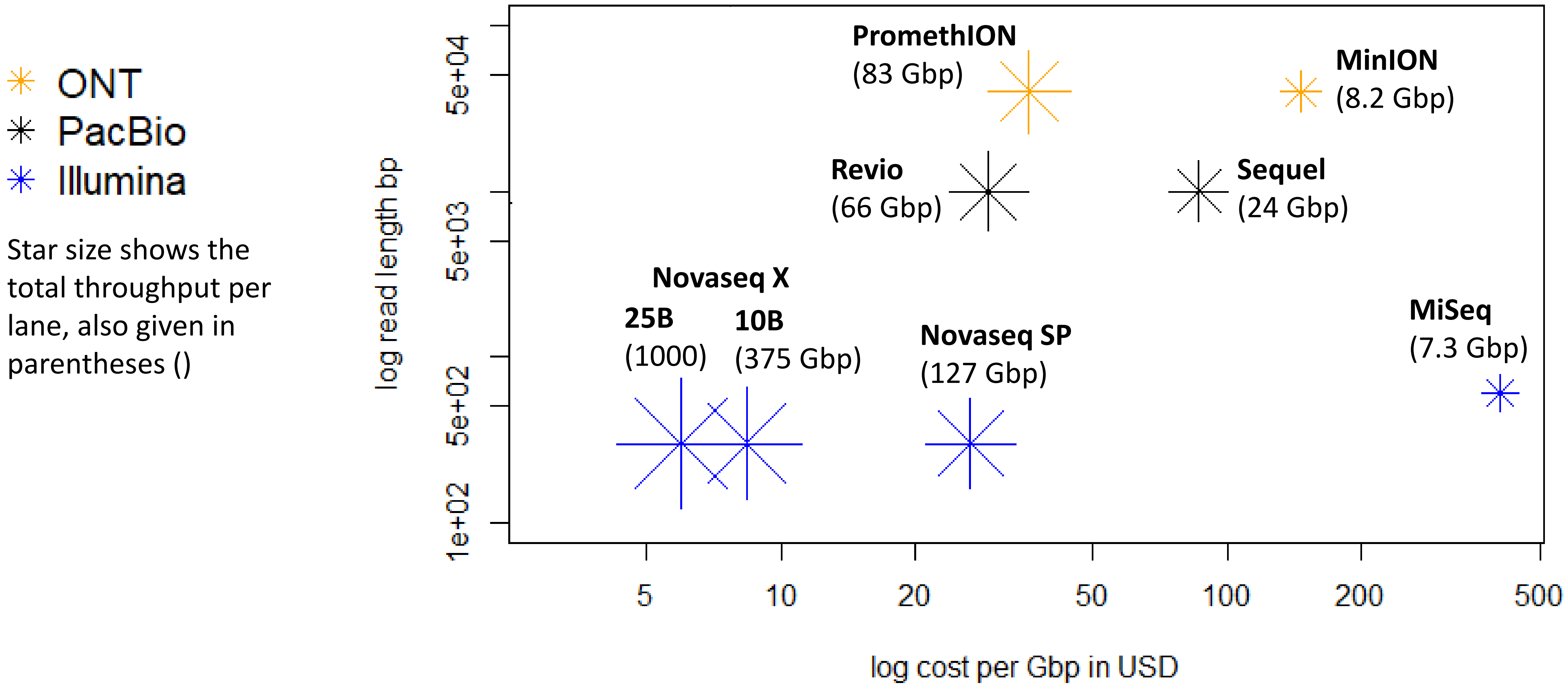
- Possible to manually check each sequence and resequence failed sequences
- Requires primer sequences and has very low throughput (expensive per bp)

Two main types of high-throughput sequencing

- **Short-read sequencing**
  - Reads are typically 150 bp long
  - Cheaper than long-read sequencing
  - E.g. Illumina, soon probably also Ultima Genomics

- **Long-read sequencing**
  - Reads are typically >10 kb long (PacBio: 15-20 kbp, Nanopore: 10-100 kbp)
  - More expensive than short-read technologies
  - Required for making a reference genome
  - E.g. PacBio or Nanopore

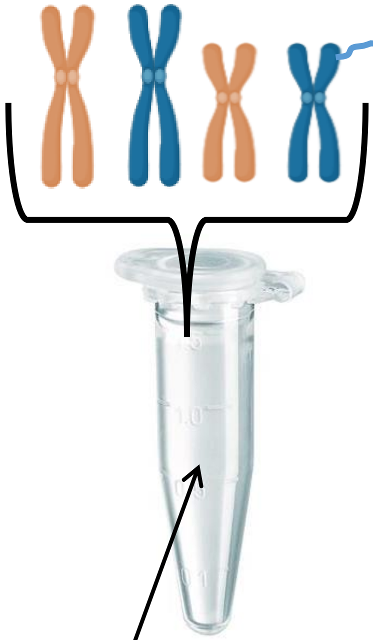
Read length versus per Gbp sequencing costs for different sequencing machines (note the axes are in logarithmic scale)



Star size shows the total throughput per lane, also given in parentheses ()

Whole-genome sequencing

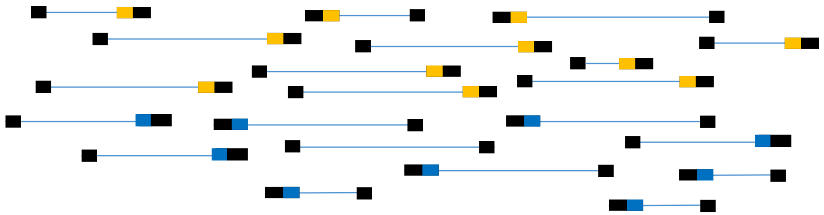
Genome
= complete set of chromosomes



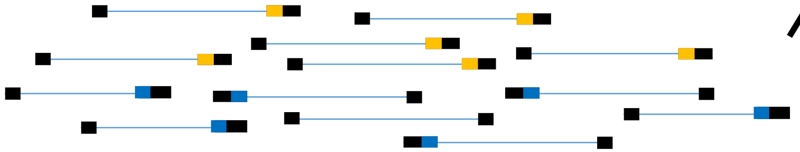
DNA (chromosomes)

Breaking chromosomes into shorter pieces for sequencing

Adding **sequencing adapters** incl. individual index



Size selection

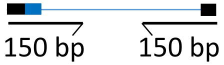


Long-read sequencing (PacBio or Nanopore/ONT)
paired-end sequencing

10-50 kbp

Short-read sequencing (e.g. Illumina)
paired-end sequencing

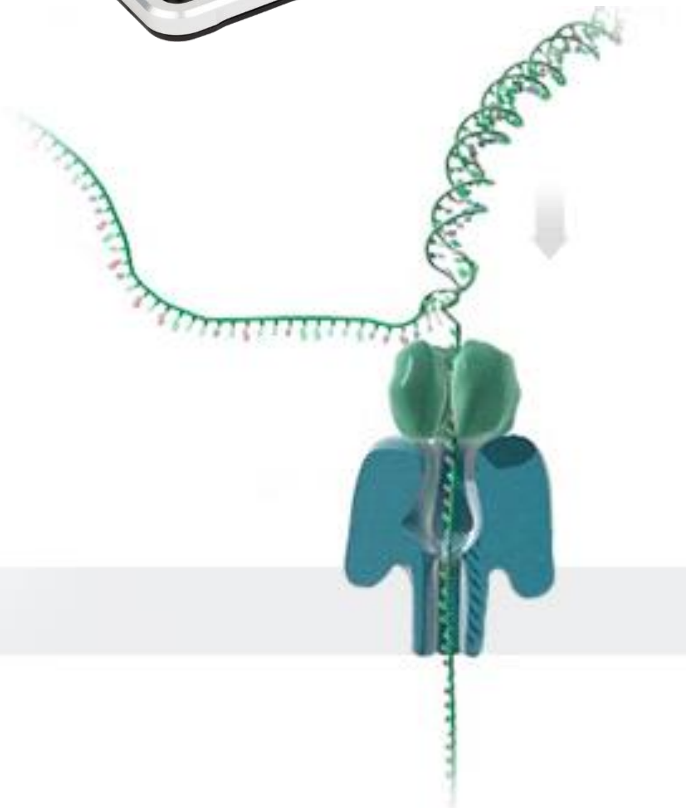
350-500 bp

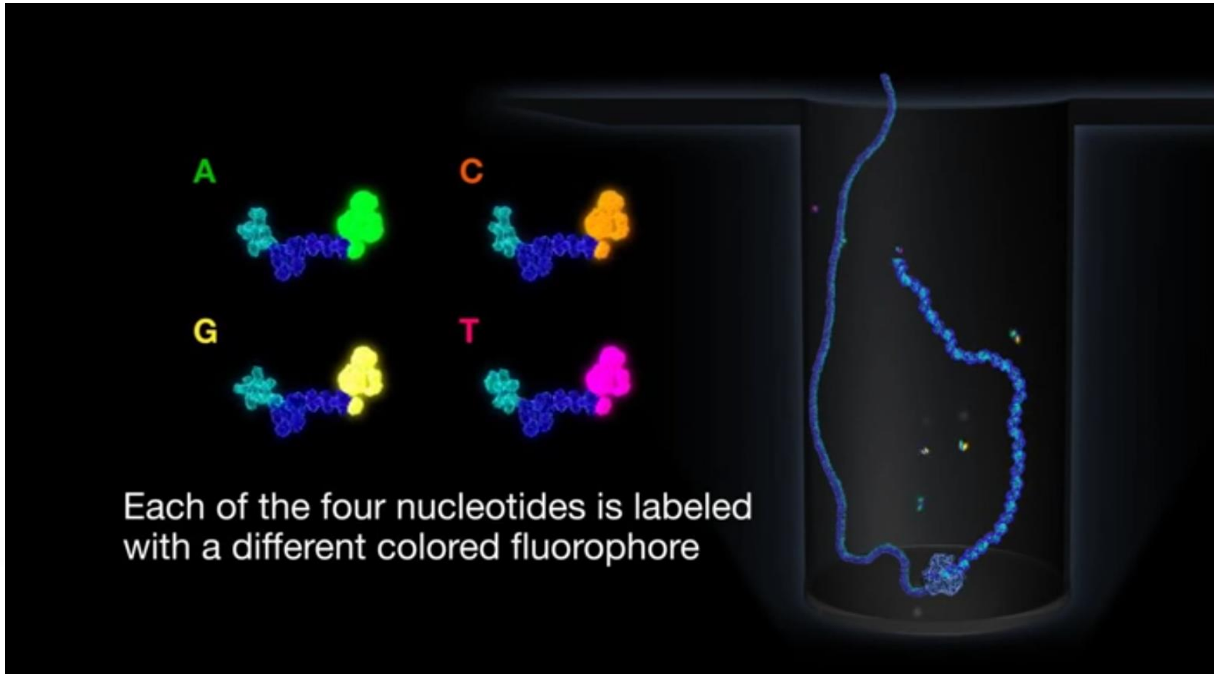


Long read sequencing technologies

Nanopore

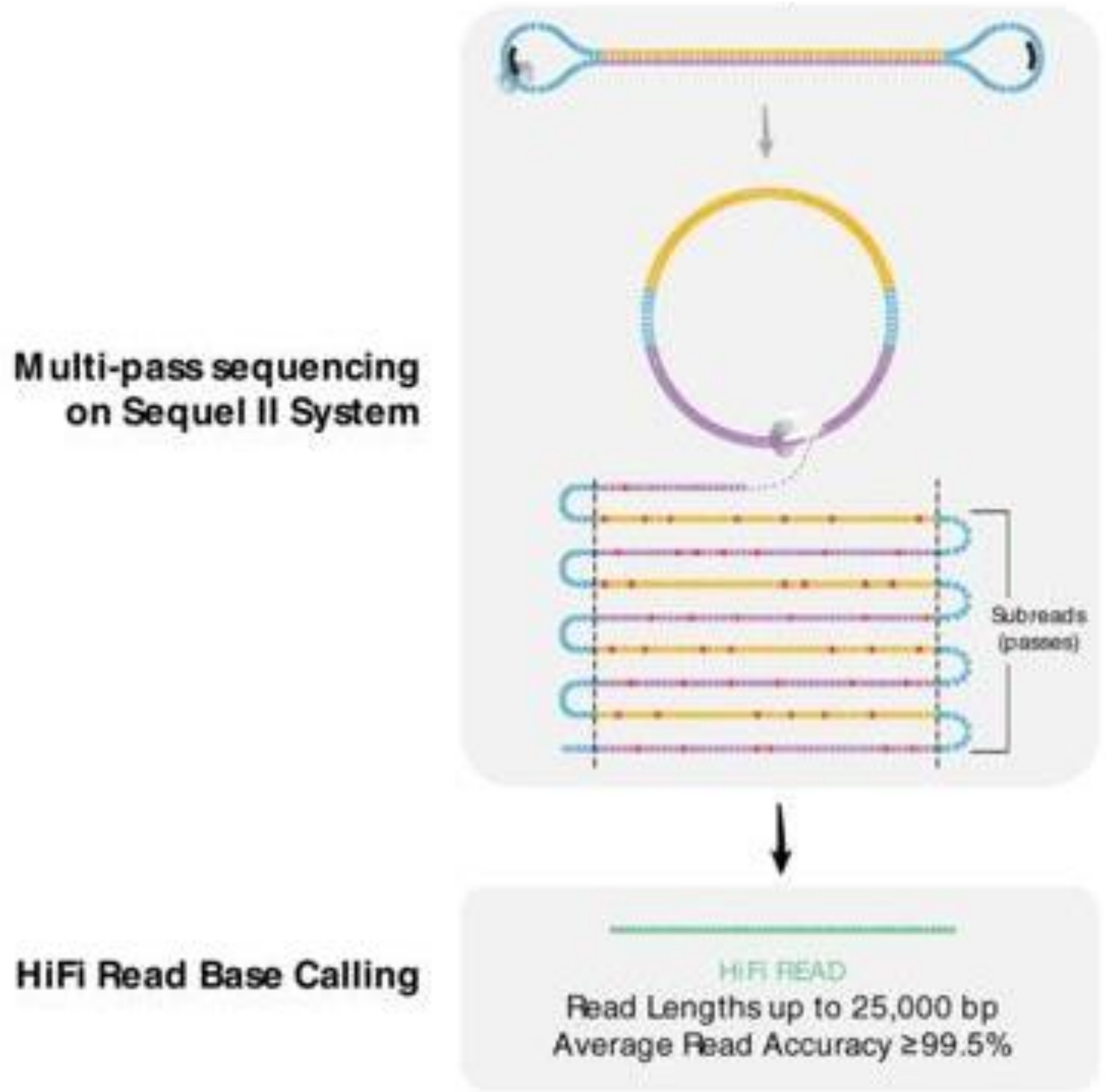
PacBio





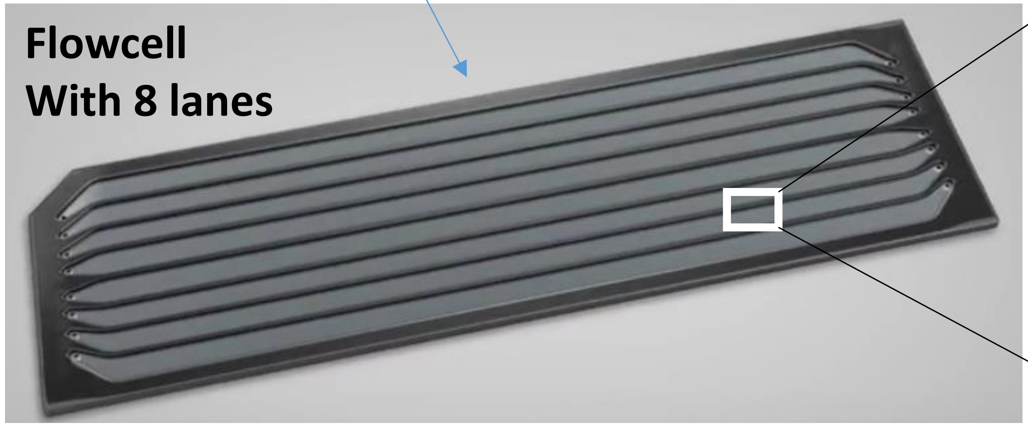
PacBio HiFi reads (99.95% accurary)

Each DNA-fragment is sequenced many times to get a high-quality consensus (=summary) read

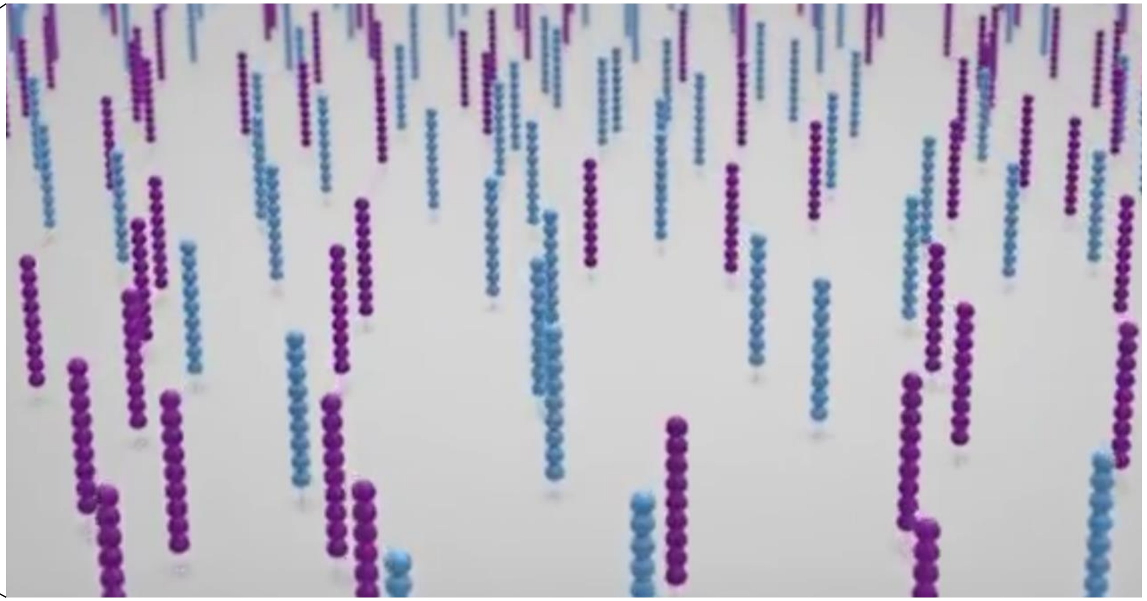


Illumina flowcell

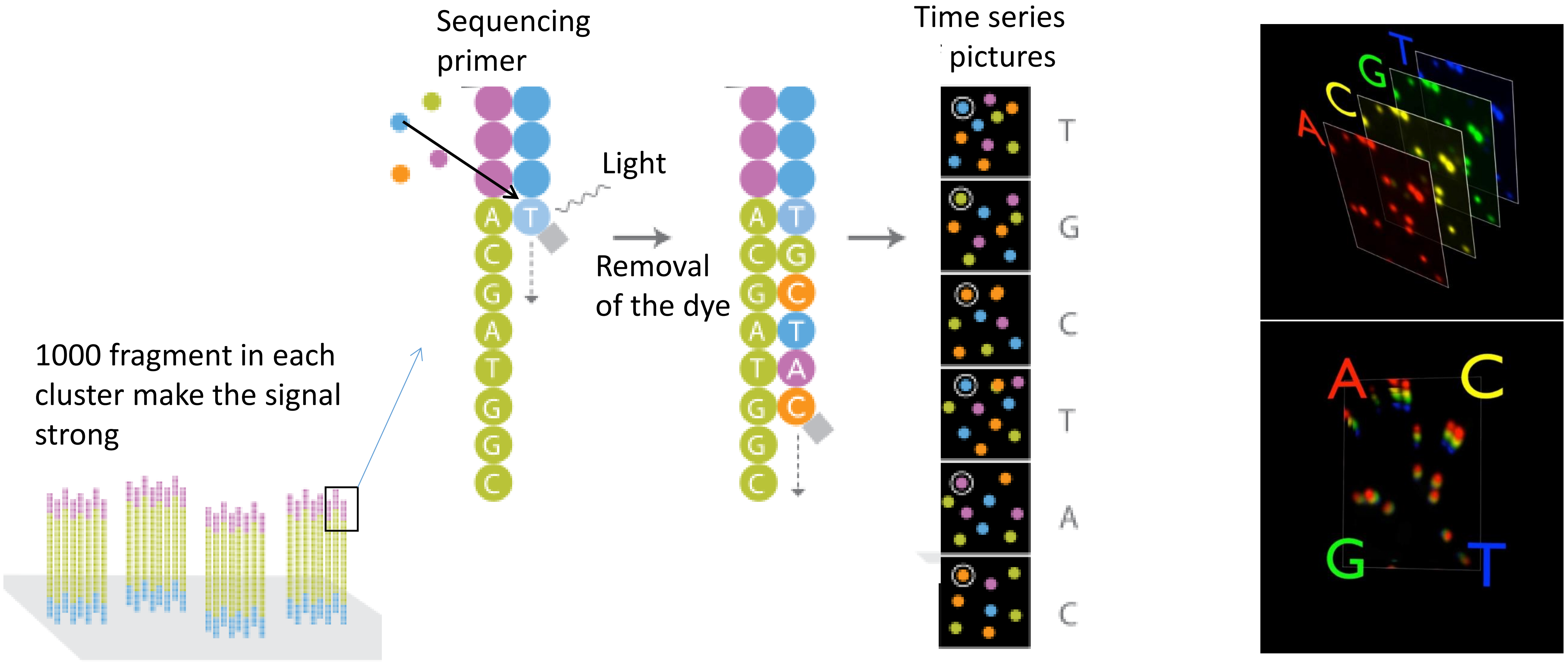
DNA fragments with Illumina adapters



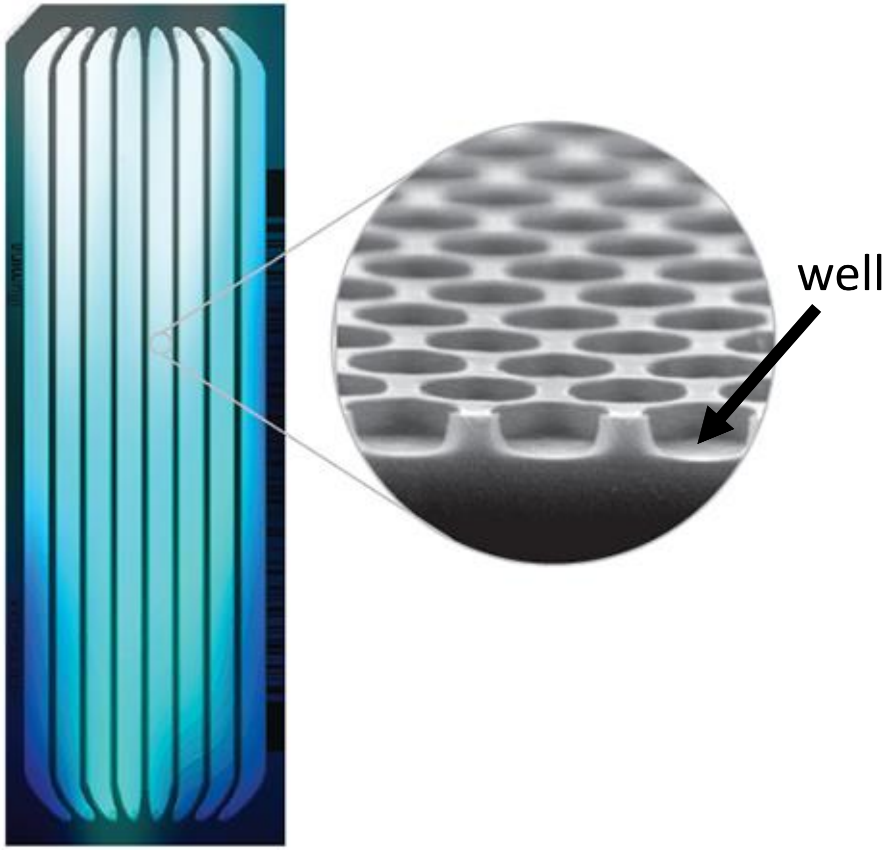
Each lane contains a dense lawn of Illumina primers

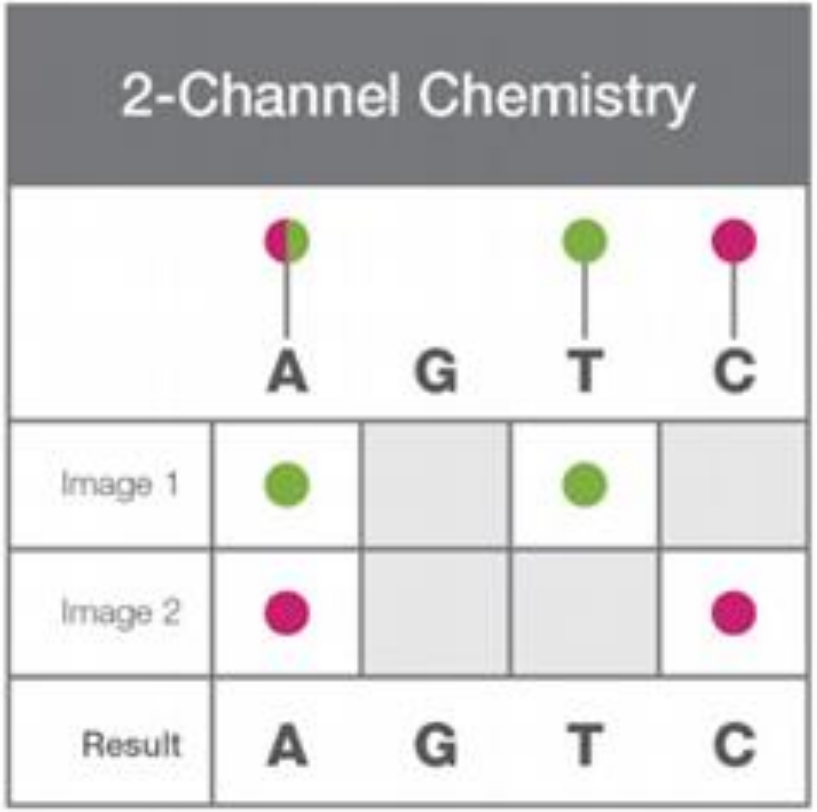


Short-read sequencing with Illumina

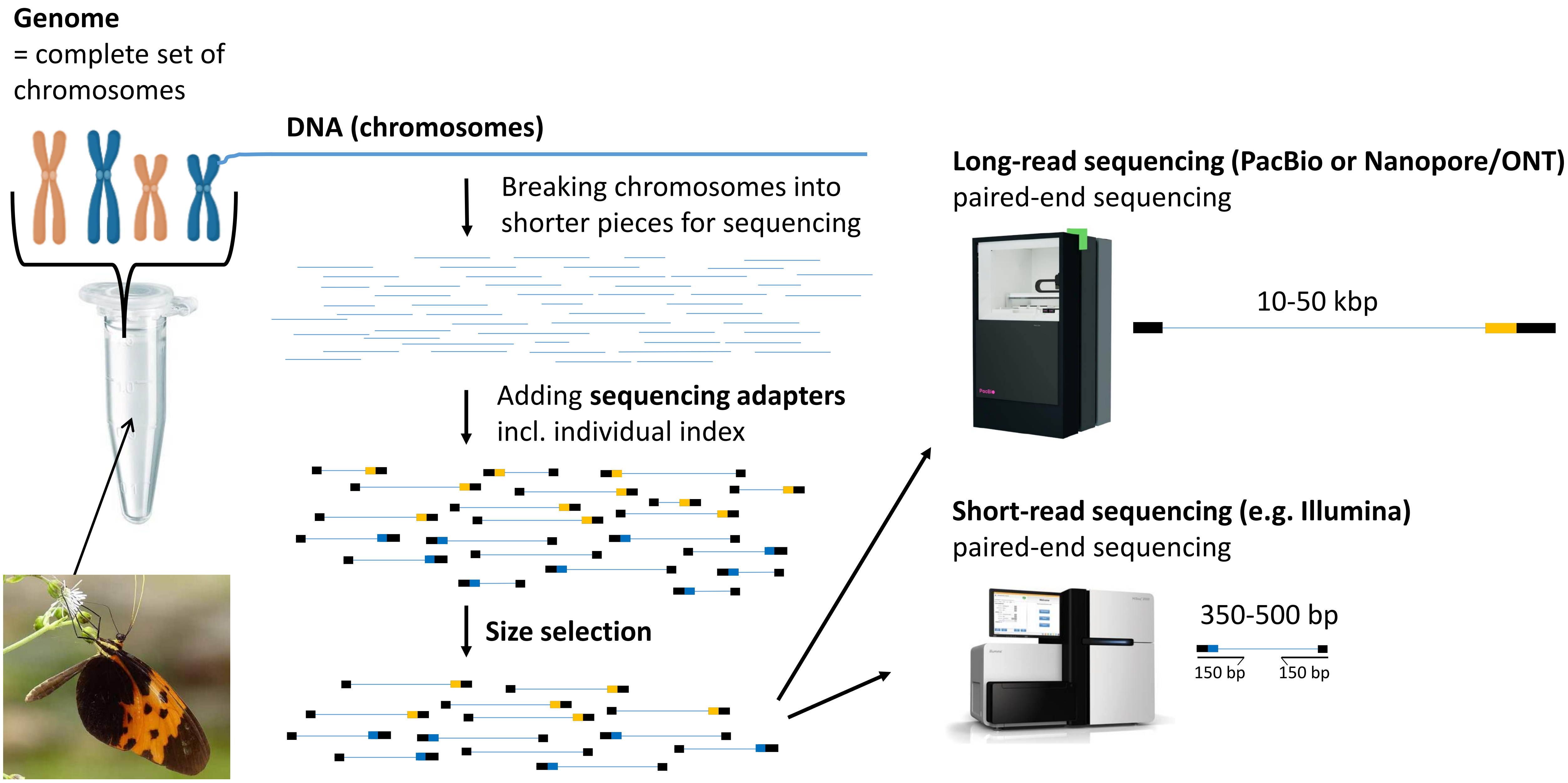


Newer Illumina machines use wells and only 2 colours
(e.g. Novaseq, Nextseq, MiniSeq. This makes it faster and cheaper)

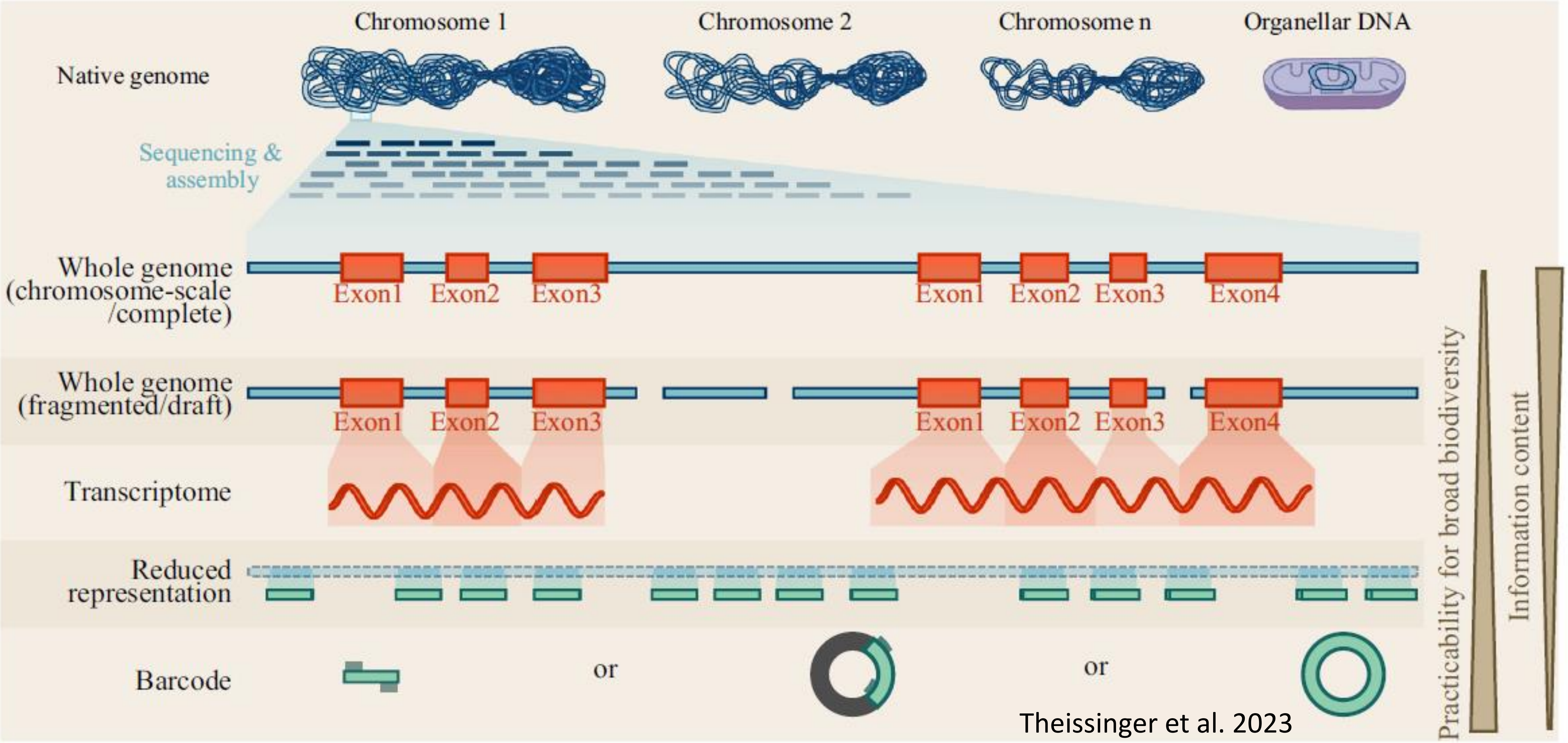




Whole-genome sequencing



Sequencing approaches for biodiversity genomics

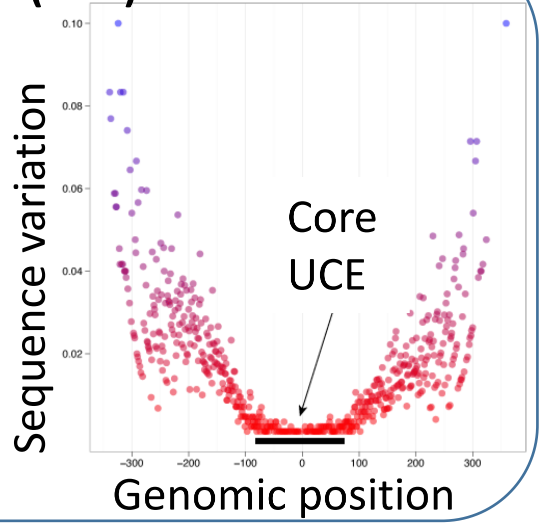


Theissinger et al. 2023

Reduced-representation techniques
(only parts of the genome sequenced)

Ultra-conserved elements (UCE)

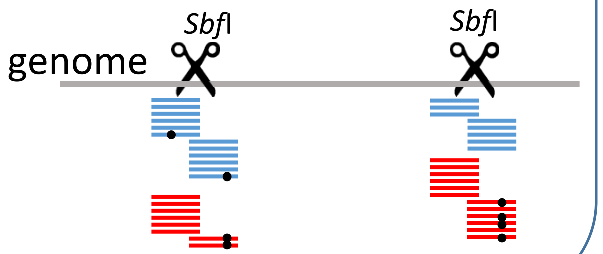
- Sequence capture with baits based on genomic regions that are conserved across many species
- Works with highly divergent species



Restriction Associated Sequencing (RAD)

(similar methods: GBS, ddRAD)

- does not require primers/baits or reference genome
- individuals need to be from the same or closely related species
- Information from thousands of loci distributed across the genome

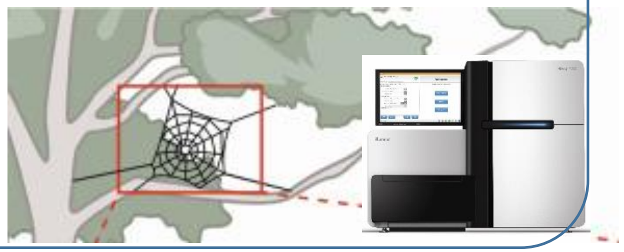


Targeted or amplicon sequencing, e.g. barcoding

- Sequencing one or few genes
- requires primers
- e.g. CO1 (mitochondrial barcoding region), advantage: large database (BOLD) available to compare to for species identification

Environmental DNA (eDNA)

- Mostly CO1 sequencing from soil, water, air (spider webs)
- Identifying local species
- Studying species richness



Trade-offs: Splitting reads (i.e. costs) among:

Total data gets divided by:

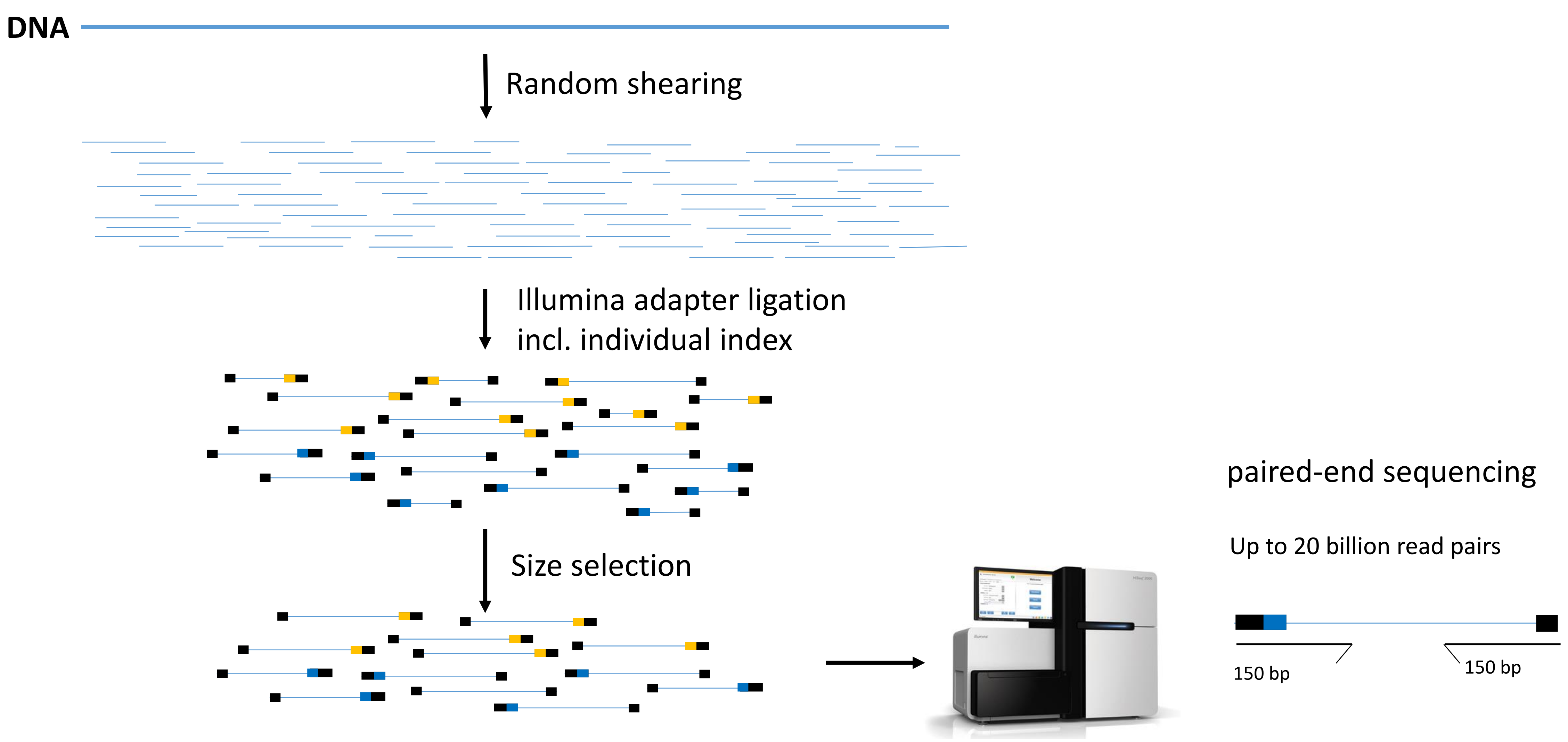
- Number of sites to sequence
  - Depends on genome size and sequencing stragegy, e.g. RAD versus whole-genome
- Sequencing depth (e.g. sequencing at 10x depth of coverage)
- Number of specimens to sequence

- Example: 1 Novaseq X 10 B lane
  ~2.5 billion paired-end reads of 150 bp each -> 375 Gbp data
  - 100 whole-genomes of a species with 0.375 Gbp genome size at 10x coverage
  - 19 whole-genomes of a species with 1 Gbp genome size at 20x coverage
  - 375 individuals sequenced with a RAD sequencing approach resulting in 50 Mbp at a sequencing depth of 20x

Now let's get started with handling genomic data!

First, a brief overview of what we will do now

Whole-genome short-read sequencing



1. Quality check and trimming raw reads

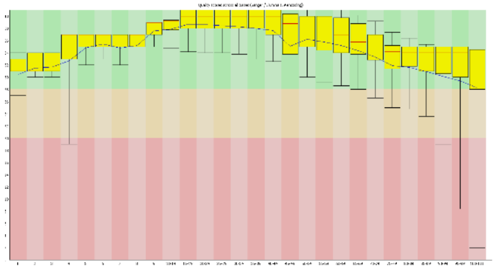
paired-end Illumina sequencing

Up to 20 billion
read pairs

Forward
reads

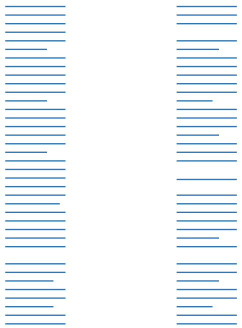
Reverse
reads

Fastqc quality plots



fastp

Trimmed and filtered
reads of good quality



Fastq file of reads

```
@HWUSI-EAS611:34:6669YAAXX:1:1:5069:1159 1:N:0:
TCGATAATACCGTTTTTTTCCGTTTGATGTTGATACCATT
+
IIHIIHIIIIIIIIIIIIIIIIIIIIIIIIHIIIIHIIIII
```

From samples to a vcf file